CREDIT SCORECARD MODEL ANALYSIS

SUMMARY

This project builds a credit scorecard derived from the well-known Lending Club dataset of US retail loans (CSV download available: https://zenodo.org/records/11295916) alongside macroeconomic data, using unpenalised and ridge logistic regression in Python.

The data is split between training and testing data, weight-of-evidence and information value scores are used to select variables, and these variables are then tested for multicollinearity using variance inflation factors. The Kolmogorov-Smirnov test statistic is used to determine the optimal threshold for classification and test validation is performed using ROC, accuracy, precision, recall, and f1 scores.

The logistic regression output is then converted into scorecard format, which is presented as an interactive Excel workbook. This gives the user the option of centering the scoring on different scores for different odds of default and observing the output given different loan amounts, loan purposes, FICO scores, etc.

To test for population drift, a holdout sample of the most recent loans is then used to compute PSI scores and the distribution of the latest loan vintage is compared to the training data using the probability quantiles from the regression model executed on the training set.

VARIABLE SELECTION

The original variables in the Lending Club dataset are given below:

Variable Name	*Data Type*	*Description*
id	Categorical	ID number for borrower
issue_d	Date (month-year)	Date when loan was issued
revenue	Numerical	Borrower's self-declared annual income in USD
dti_n	Numerical	Debt to income ratio
loan_amnt	Numerical	Amount borrowed in USD
fico_n	Numerical	Borrower's FICO score
experience_c	Boolean	Whether the borrower has existing loans with the entity or not (0 = no, 1 = yes)
emp_length	Ordinal	Employment length in years (NI = no information)
purpose	Categorical	Funding purpose of the loan
home_ownership_n	Categorical	Home ownership status: mortgage, rent, own or other
addr_state	Categorical	Borrower's state code
zip_code	Categorical	Borrower's ZIP code (last two digits omitted)
Default	Boolean	Whether the borrower defaulted or not (0 = no, 1 = default)
title	Categorical	Title of the credit request description provided by the borrower

desc	Textual	Description of the credit request

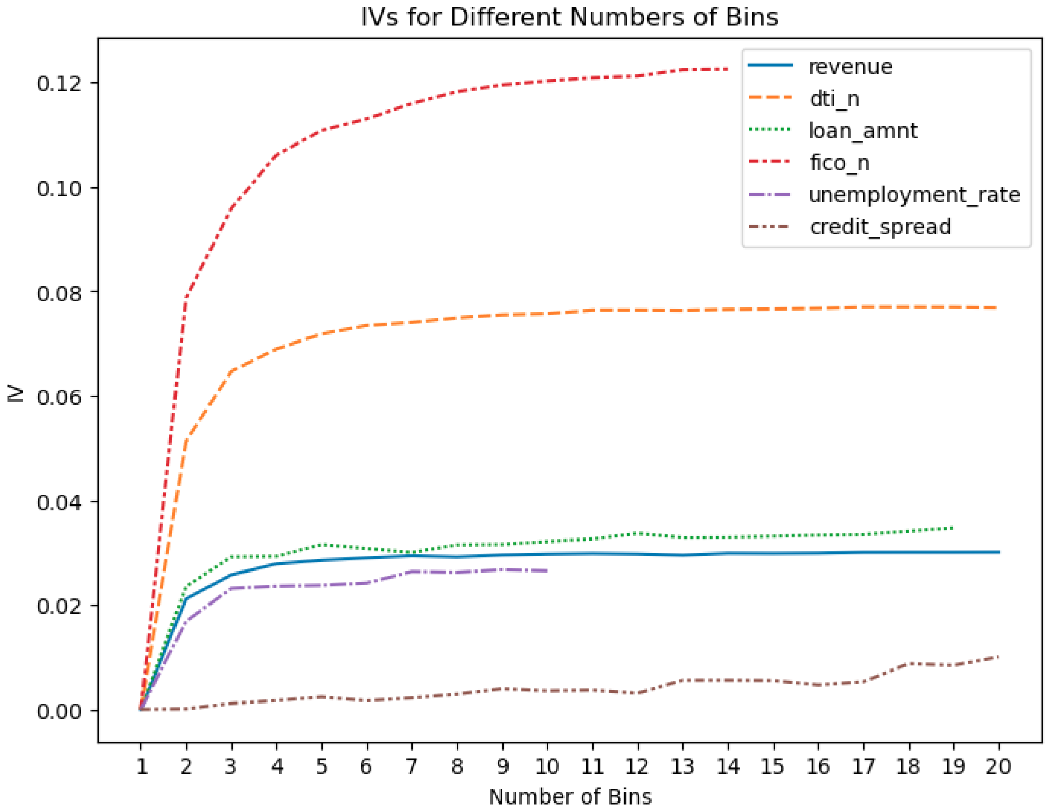
To these original variables, the following were added as proxies for the state of the economy on the loan origination date:

Variable Name	*Data Type*	*Description*
unemployment_rate	Numerical	US monthly unemployment rate (%) at issue_d, lagged by a month to take account of lagged data reporting. Source: ALFRED (to eliminate data revisions).
credit_spread	Numerical	Average monthly US BBB corporate credit spread for month of issue_d. Source: FRED

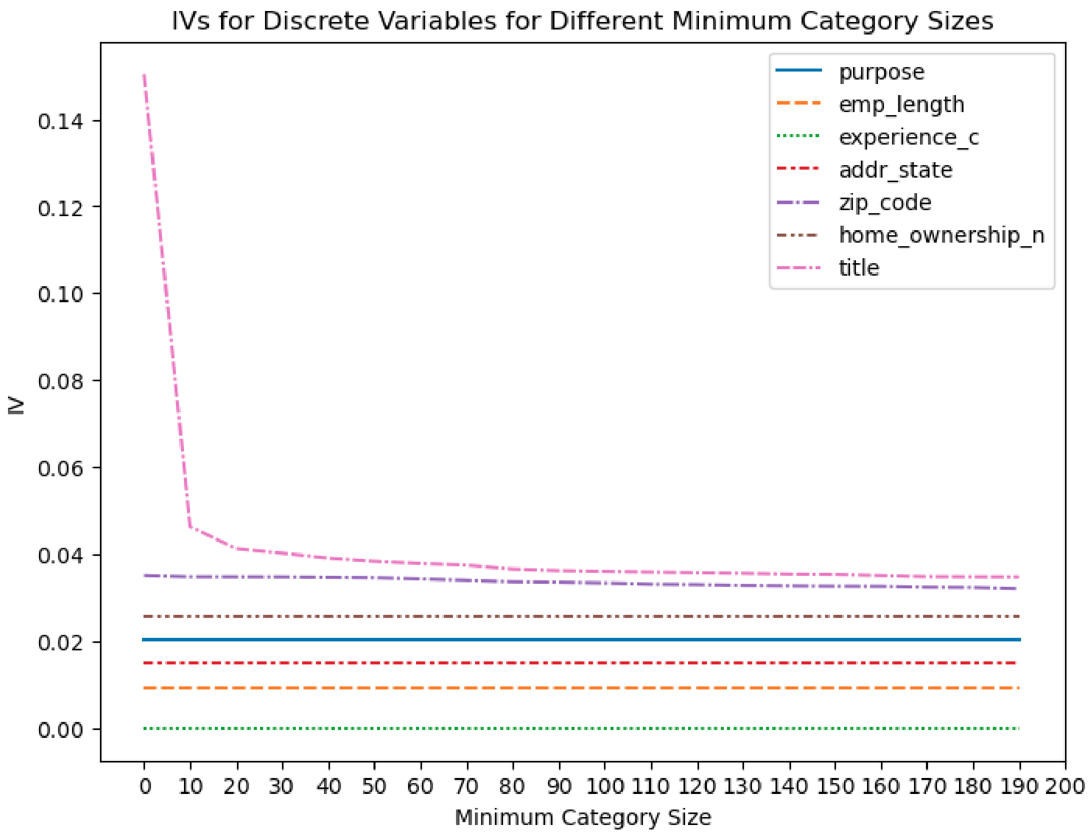
The following variables were initially dropped for the regression for the following reasons:

Variable Name	*Reason for Dropping*
id	No relevance for default probability.
desc	Insufficient data.

From the remaining variables, information values were calculated from weight-of-evidence scores with different numbers of percentile bins tested for the continuous variables:



Since some of the categorical variables (e.g. title) contain many different minority entries, IVs for the discrete variables were computed at different minimum thresholds for category sizes:

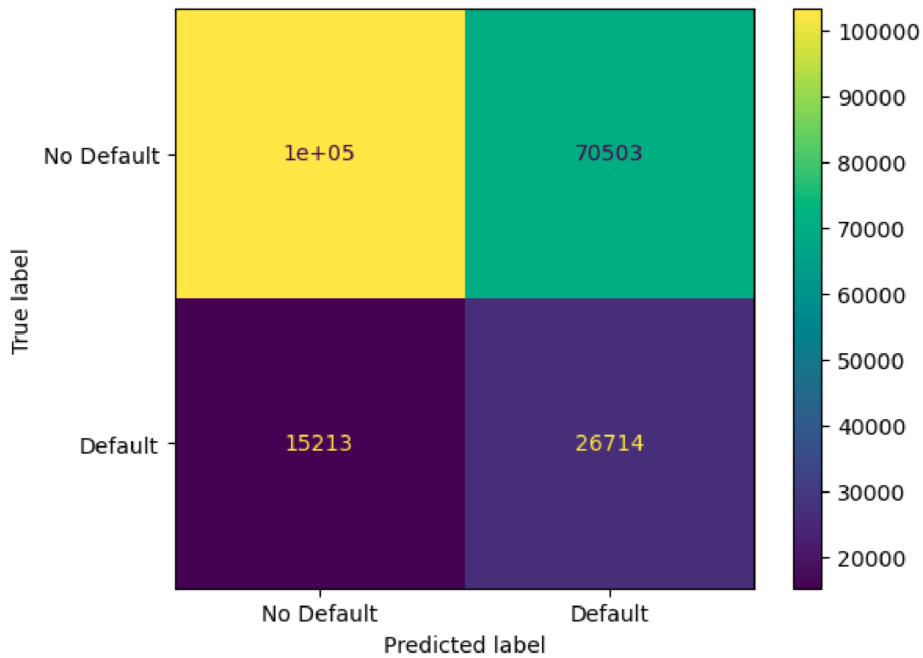


Due to low IV values (less than 0.02), creditspread, emp_length and addr_state were subsequently dropped for the regression stage. 10 bins were chosen for the continuous variables as this was the maximum number of possible equal-percentile-width bins for the unemployment_rate variable (without hitting plus or minus infinity for the weight-of-evidence score due to homogeneous scores within bins) and the other variables level off with low marginal IV gains after 10 bins anyway.

A minimum category size of 20 was chosen to nullify the inflated effects of training on many different entries for the title variable, without losing informative information. Categories with less than 20 entries were rewritten as 'Other'.

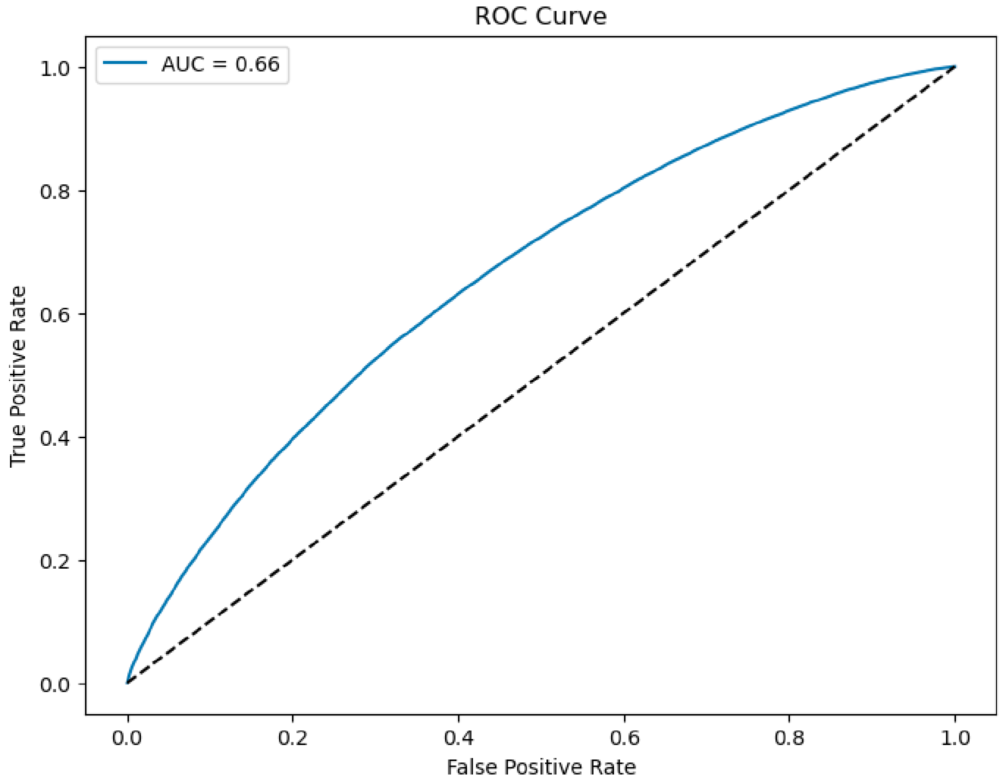
LOGISTIC REGRESSION

The data was split between training and testing data in the ratio 80%:20% and the Default=1 class subset was overweighted as the minority class by choosing the "balanced" weighting parameter for logistic regression. Each dataset entry was replaced with bin weight-of-evidence scores and the K-S statistic was used to determine the optimal classification threshold. The model coefficients, ROC, confusion matrix and validation scores from evaluation on the test sample are below:




```
Precision: 0.27478733143380274
Accuracy: 0.6024838959509157
Recall: 0.6371550552150166
F1 Score: 0.3839763123095498
```

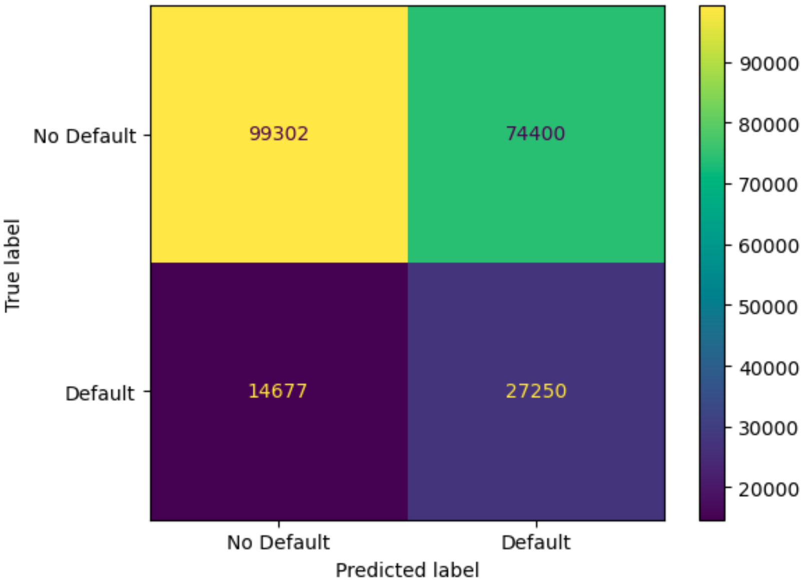
```
            Variable  Coefficient
0          intercept     0.049427
1            revenue    -1.210339
2              dti_n    -0.672014
3          loan_amnt    -1.883166
4             fico_n    -0.962888
5            purpose    -0.564616
6    home_ownership_n   -0.881843
7           zip_code    -0.929684
8              title    -0.396594
9   unemployment_rate   -0.610636
```



Independent variables were tested for multicollinearity using variance inflation factors, but these returned as normal (the maximum implied R^2 value equalled 0.4312, from the title variable):

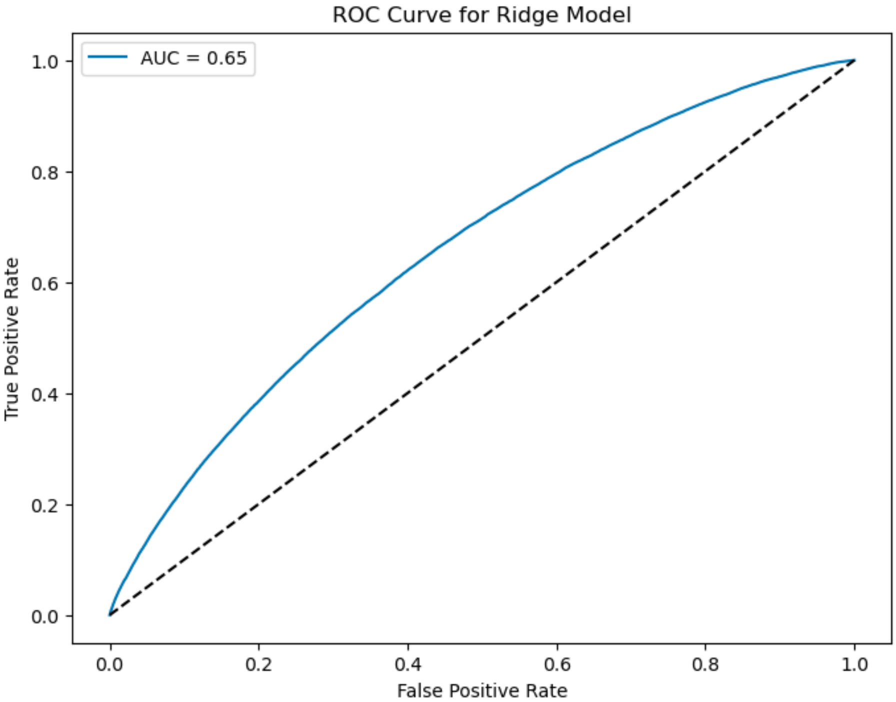
```
title                 1.757978
purpose               1.499984
revenue               1.412546
loan_amnt             1.285725
unemployment_rate     1.272872
dti_n                 1.096659
home_ownership_n      1.094905
fico_n                1.033574
zip_code              1.015337
dtype: float64
```

To ensure that the out-of-sample performance of the model could not be improved by reducing the model complexity, a ridge logistic regression model was also trained with a heavy penalisation parameter (C=0.0001). The results of this are below:



	Variable	Coefficient
0	intercept	0.024191
1	revenue	-0.309709
2	dti_n	-0.525855
3	loan_amnt	-0.478872
4	fico_n	-0.674175
5	purpose	-0.196374
6	home_ownership_n	-0.305956
7	zip_code	-0.391141
8	title	-0.329959
9	unemployment_rate	-0.228342

```
Precision: 0.2680767338908018
Accuracy: 0.5868969387234556
Recall: 0.6499391800033392
F1 Score: 0.3795872597978785
```



Penalising the regression coefficients results in lower validation scores across the board, and (unusually for a ridge model), does not result in fewer effective independent variables. This indicates that all the variables used have predictive power and dramatically reduce model effectiveness if removed. Given the variables included in the regression, the model is optimal.

Both models have accuracy and recall scores better than chance (0.5) but poor precision scores, reflecting a large proportion of false positives. The K-S threshold choice to maximise the gap between the true positive rate and false positive rate is designed to optimise recall, which will inevitably reduce precision. Given that, in this scorecard context, a false positive (predicting a default that does not occur in the end) is less problematic than a false negative (not warning that a borrower is about to default), optimising for recall is preferable.

The poor precision score may also be modestly alleviated by not overweighting the defaulting class, but at the expense of a lower recall score.

CREDIT SCORECARD CONVERSION

An interactive Excel workbook is provided with this project which converts the output of the unpenalised logistic regression into scorecard format. Below is a sample screenshot:

BASE SCORE	BASE ODDS OF DEFAULT	POINTS TO DOUBLE SCORE	DEFAULT LOG-ODDS	CREDIT SCORE	REVENUE (UPPER BOUNDS)		TOTAL DEBT / MONTHLY INCOME (UPPER BOUNDS)		LOAN AMOUNT (UPPER BOUNDS)		FICO SCORE (UPPER BOUNDS)		LOAN PURPOSE		HOME OWNERSHIP		ZIP CODE	
600	50	20	0.0759	485	Enter:	50000	Enter:	15	Enter:	10000	Enter:	700	Choose:	debt_consolidation	Choose:	RENT	Choose:	211xx
					VALUE:	0.1519	VALUE:	-0.1156	VALUE:	-0.2618	VALUE:	-0.0828	VALUE:	0.0437	VALUE:	0.1529	VALUE:	0.0547
					34000	0.2708	7.39	-0.2682	5000	-0.5171	667	0.3324	small_business	0.2673	RENT	0.1529	415xx	1.0068
					42000	0.1519	10.56	-0.2283	7000	-0.4247	672	0.2318	moving	0.1063	OTHER	0.1137	738xx	1.0024
					50000	0.1519	13.07	-0.1659	9000	-0.4247	677	0.1912	renewable_energy	0.0991	OWN	0.0299	736xx	0.9849
					56000	0.0699	15.36	-0.1156	10000	-0.2618	682	0.1053	house	0.0903	MORTGAGE	-0.1453	833xx	0.7604
					65000	0.0699	17.62	-0.0551	12000	0.0366	692	0.1053	educational	0.0658			739xx	0.7093
					73000	0.0132	19.96	-0.0098	15000	0.1772	697	-0.0828	medical	0.0598			583xx	0.6850
					84000	-0.0744	22.56	0.0460	18000	0.1772	707	-0.0828	debt_consolidation	0.0437			689xx	0.6822
					100000	-0.2835	25.6	0.1099	21000	0.2399	717	-0.3960	other	0.0399			638xx	0.6667
					125000	-0.4116	29.66	0.1933	27350	0.3048	737	-0.3960	vacation	-0.0255			686xx	0.6590
					9550000	-0.4116	999	0.3424	40000	0.3476	847.5	-0.8008	home_improvement	-0.0756			438xx	0.6542
													major_purchase	-0.0795			683xx	0.6318
													credit_card	-0.1189			687xx	0.6263
													car	-0.2139			669xx	0.6121
													wedding	-0.2999			464xx	0.5594
																	749xx	0.5381
																	237xx	0.5296
																	639xx	0.5275
																	753xx	0.5275

The user can select a base score with odds of default to center the model, alongside the number of points needed for the score to double (20 is a common value for the latter). Dropdown menus are used to choose settings for the categorical variables, and continuous variables can be entered, so long as the entries are in the range of the training data.

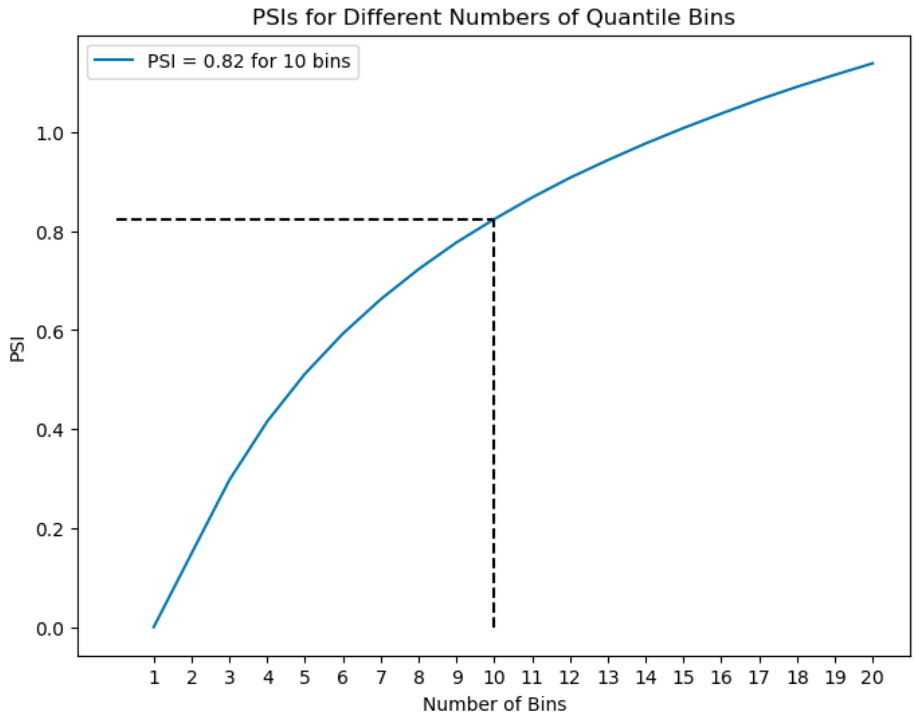
The conditional formatting of each variable provides useful feedback on which settings might lead to greater probability of default:

Variable	*Discussion*
Revenue	• Unsurprisingly, there is a strong negative correlation between annual income and probability of default.
Total Debt / Monthly Income	• Likewise, there is an unsurprising positive correlation between the debt ratio and default rates.

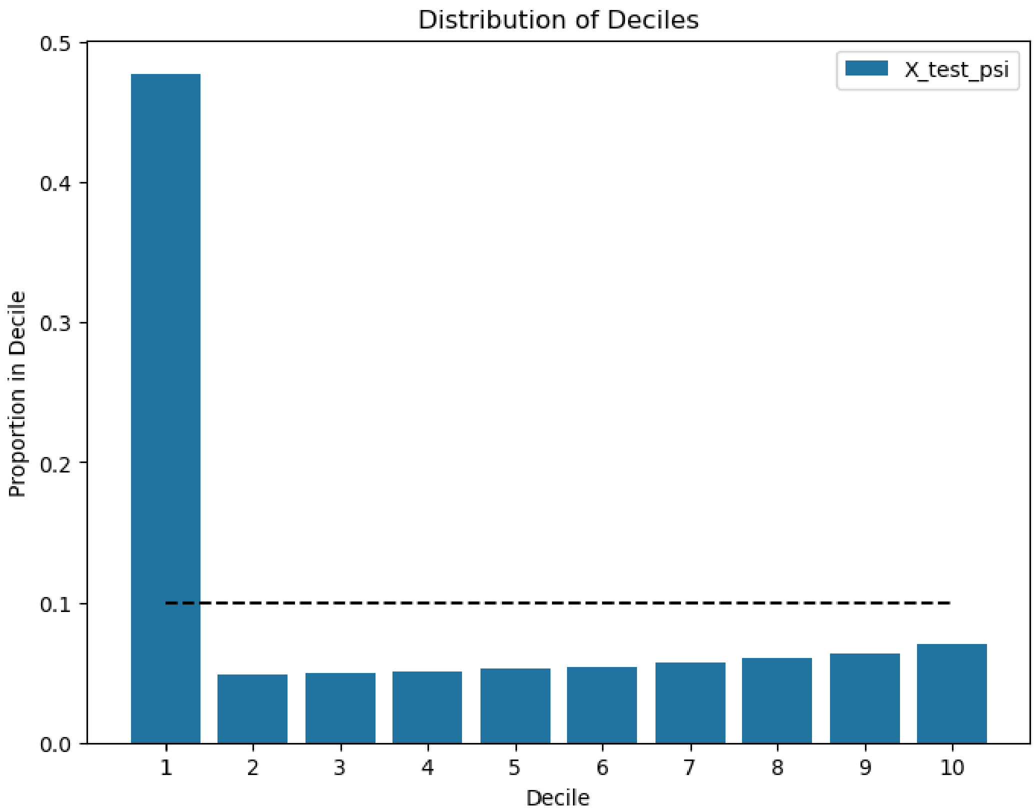
Loan Amount	• Similarly, the greater the loan amount, the greater the default probability.
FICO Score	• Unsurprisingly, there is a strong negative correlation between FICO score and default risk.
Loan Purpose	• Wedding and car loans are the least risky; • Small business loans are by far the riskiest.
Home Ownership	• Renters are the most risky; • Mortgage owners are the least risky, surprisingly less risky than those that own their home outright.
Zip Code	• From matching the zip codes to geographical regions: • The three lowest risk zip codes are in Wyoming, Massachusetts and Maine • The three highest risk zip codes are in Kentucky and Oklahoma (x2)
Loan Title	• Further key word analysis is needed to determine the relationship between key words and model scores for this variable, possibly using an unsupervised ML model.
Unemployment Rate	• Interestingly, there is a negative relationship here between the unemployment rate and probability of default: • This could be explained by timing: the US unemployment rate is taken from loan origination (lagged by a month to take account of data reporting lags), so an economy near the peak/trough of the business cycle may be more/less likely to deteriorate throughout the loan period leading to a greater/lesser probability of issuer default.

PSI ANALYSIS

The most recent 20% of loans were held out to compute PSI values for various bin numbers, using quantile bins for predicted probabilities from the unpenalised model on the data used for model training. The results of this are below (including the value for the industry standard of 10 bins):

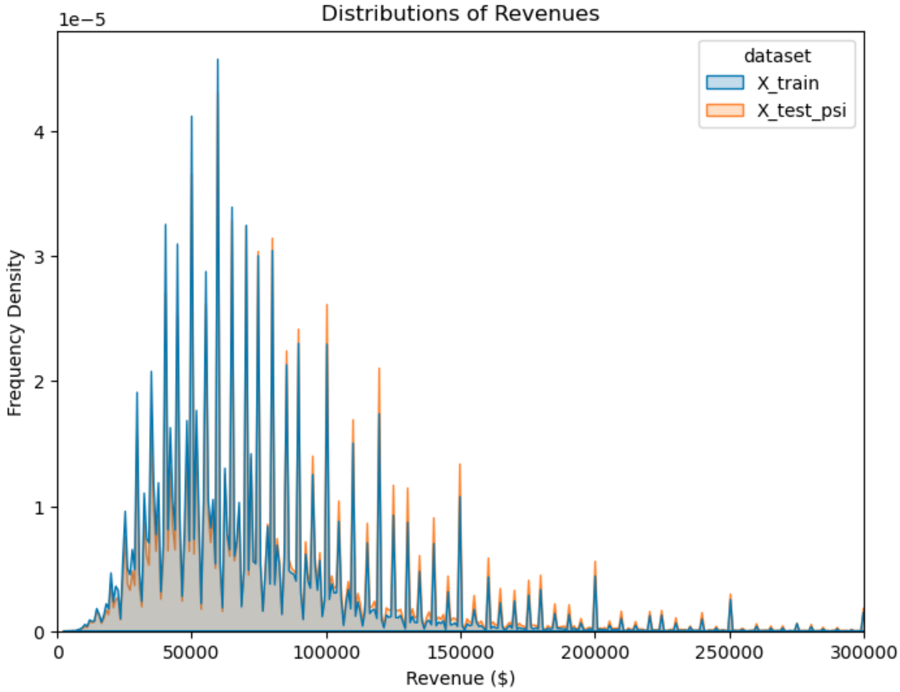


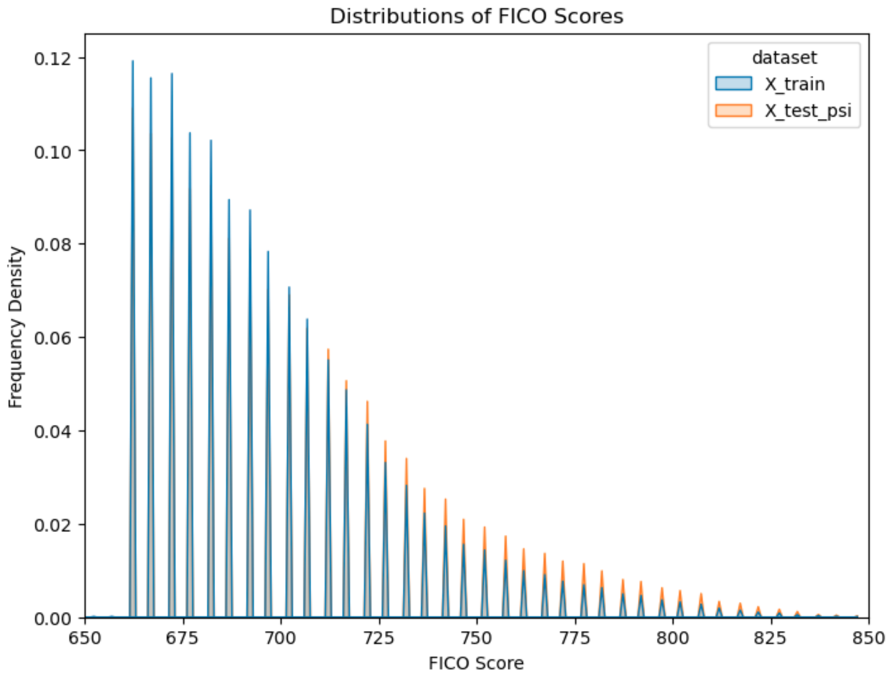
0.82 is a very high value, indicating a shift in default probabilities in the underlying loan population for the later vintage. To see this shift graphically, the distribution of proportions of defaults in decile bins is given below. A horizontal line is added to the chart for reference, which indicates the shape if there were no underlying shift in the default distribution:

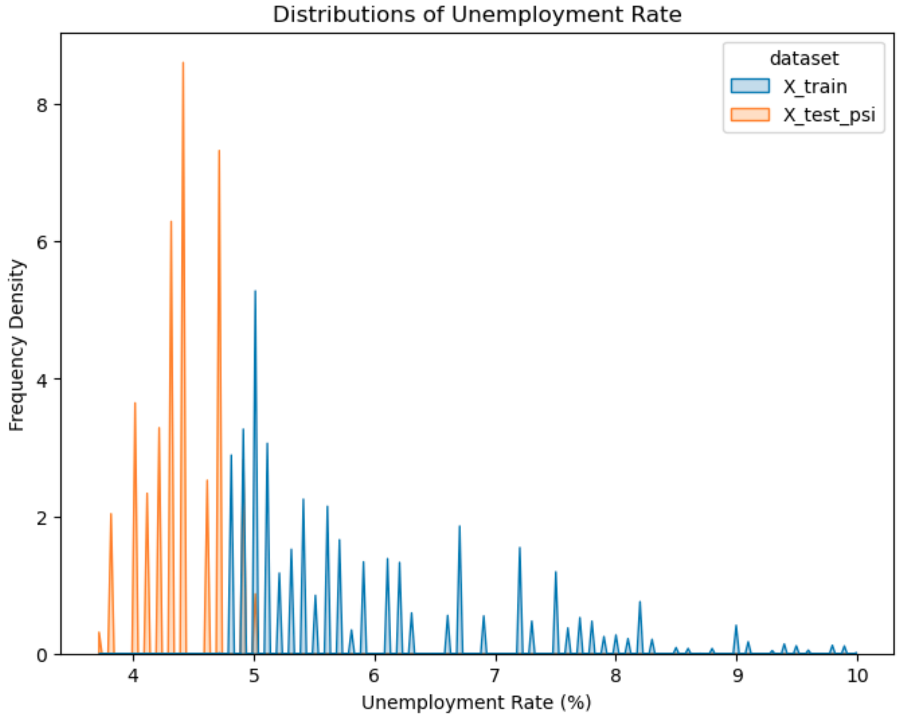


This demonstrates that the shift in the latest vintage is towards much lower probabilities of default. It should be said that this is not simply due to the latest loans being yet to mature: all loans in the Lending Club dataset were filtered to only include loans either completed (without default) or defaulted.

To examine this discrepancy further, below are graphs of the distributions for each variable used by the model (except title), restricted to those which show differences between the training and holdout sets:







The debt-to-income ratio, home ownership type, loan amounts, loan purposes and zip codes showed insufficient distributional differences likely to substantially affect the shift in default probabilities.

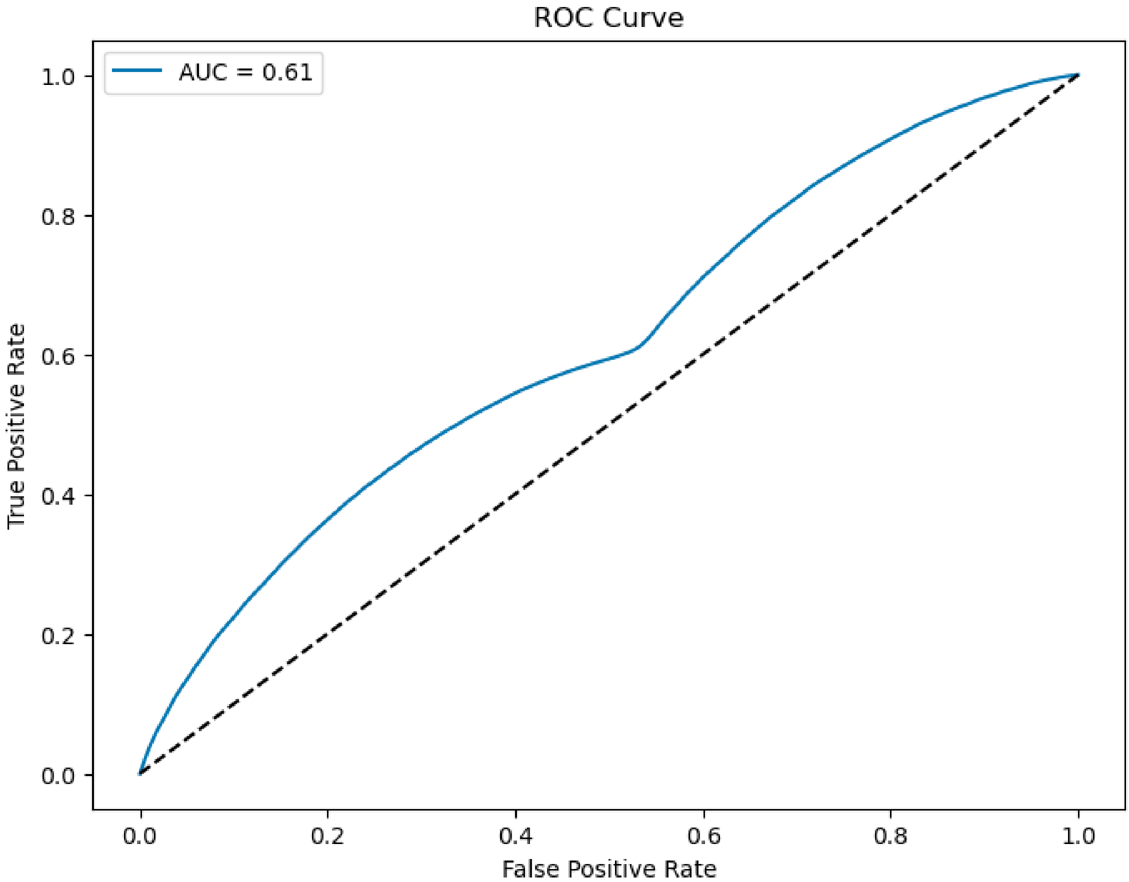
From the graphs, a slight shift upwards in the distribution of revenues and FICO scores would have reduced the likelihood of default in the holdout sample, although this would have been offset by the decline in the unemployment rate (which, as it turns out, negatively impacts the output credit score for reasons already explained).

The remaining variable to consider is the title variable. New entries in the holdout sample with unrecognised labels were assigned as ‘Other’, with a score of model coefficient times weight-of-evidence as -0.0596, which is negative. Given the variety of titles, it is likely that new labels were abundant in the holdout same, and the resulting negative weight-of-evidence shift from these labels led to a downward shift in the probability of default.

The combined reduction in probabilities of default from revenue, FICO score and the treatment of the title variable were apparently enough to offset the reduction in the unemployment rate, leading to much lower rates of default.

There is a possibility that the apparent shift in distribution is due to model performance on the new data. From the screenshots below, there was an improvement in the precision score and a decrease in the recall score, which indicates that the model produced fewer false positives (flagged as defaulting when no default occurred) relative to true positives and more false negatives (flagged as not defaulting when default occurred) relative to true positives on a proportional basis, with an increase in accuracy and a slight reduction in F1 score. This gives a clearer reason for the downward shift in default probabilities on the new data.

```
Precision: 0.31384278993206097
Accuracy: 0.6697509794609996
Recall: 0.4352099480453113
F1 Score: 0.36469395911841956
```



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